Rapport diagnostic des parametres alpha - DMM et DPMM

Objectif du rapport

Ce document verifie si les composantes estimees par les modeles de melange de Dirichlet presentent des parametres raisonnables. Il ne sert pas a choisir K directement; il sert a documenter la non-degenerescence globale des profils.

Definitions

Pour une composante c et J postes budgetaires, le vecteur alpha_c = (alpha_c1, ..., alpha_cJ) definit le profil moyen E(w_cj) = alpha_cj / alpha_0c, avec alpha_0c = somme_j alpha_cj. Un alpha_0 plus eleve indique une composante plus concentree autour de son profil moyen. Les poids pi, en DMM, et posterior_weight_mean, en DPMM, indiquent l'importance effective des clusters ou composantes.

Perimetre

DMM: K = 2 a K = 9 pour 2007/2014 et 9/23 postes. DPMM: Kmax = 7 pour les quatre priors disponibles et les quatre cas.

Lecture pratique: chercher simultanement des alpha0 coherents et des poids non-negligeables. Un alpha_j individuel faible peut simplement correspondre a un poste peu representé dans un profil; ce n'est pas, seul, un probleme.

Synthese globale du diagnostic alpha

Lecture: alpha0 est la somme des parametres alpha_j d'une composante de Dirichlet. Il donne un diagnostic de concentration interne du profil moyen. Des valeurs tres faibles, combinees a des poids de clusters minuscules, signaleraient des composantes peu fiables.

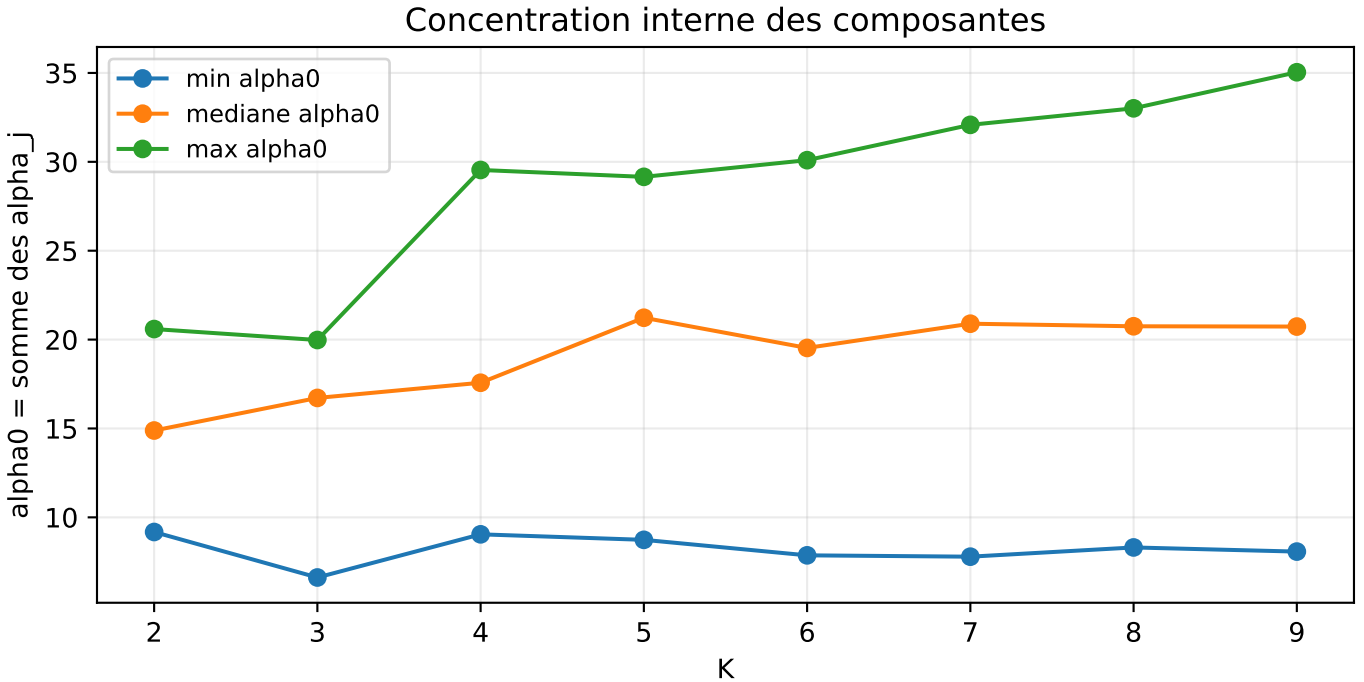
Conclusion: les solutions estimees ne montrent pas de degenerescence evidente. Les alpha0 restent dans des plages coherentes, tandis que les poids de composantes permettent d'identifier les profils substantiels. Ce diagnostic complete les criteres de probabilite d'affectation, d'ambiguite et de taille minimale; il ne choisit pas K a lui seul.

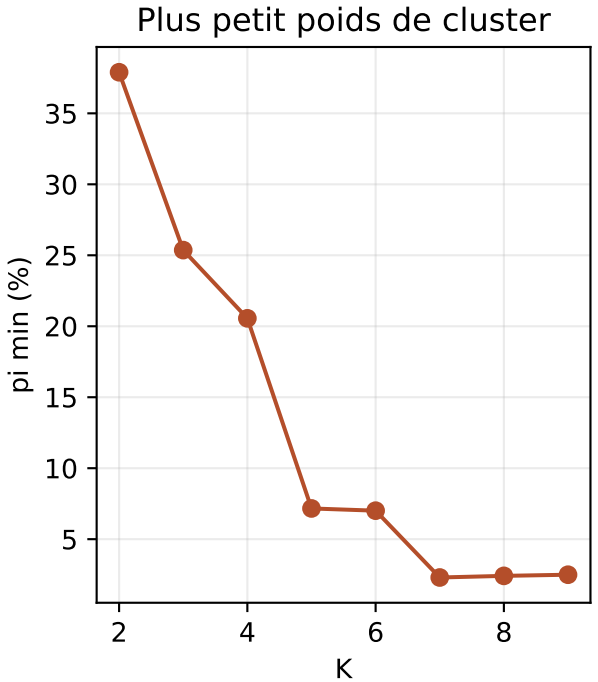
Modele	Base	alpha0 min	alpha0 med.	alpha0 max	poids min %	alpha_j min	alpha_j max
DMM	32 solutions K-cas	6.08	20.47	35.04	1.01	0.092	17.43
DPMM	16 prior-cas	5.64	21.52	47.66	0.98	0.089	23.31

Note: dans le DPMM, les labels gamma_* designent le prior sur le parametre de concentration du processus de Dirichlet. Ils ne doivent pas etre confondus avec les alpha_j des profils de consommation.

Fichiers CSV generes: diagnostic_alpha_dmm_resume.csv et diagnostic_alpha_dpmm_resume.csv

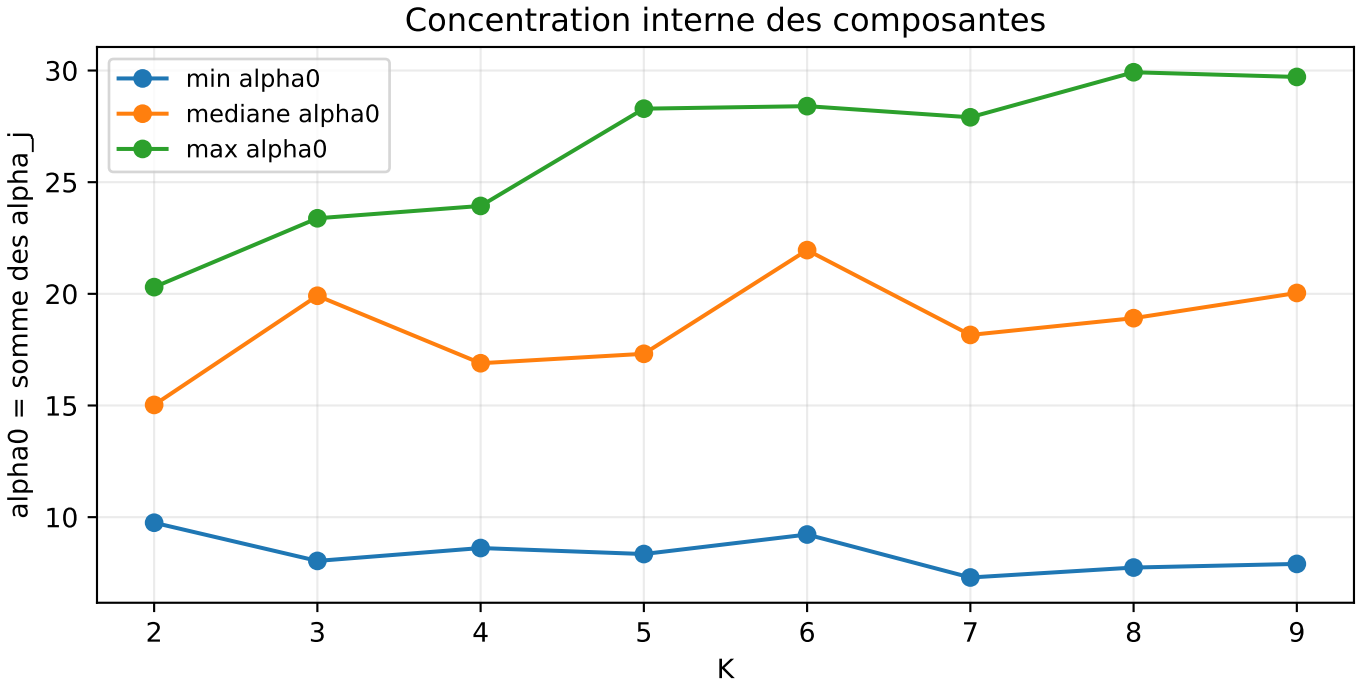
DMM - Diagnostic alpha0 - 2007 - 9 postes

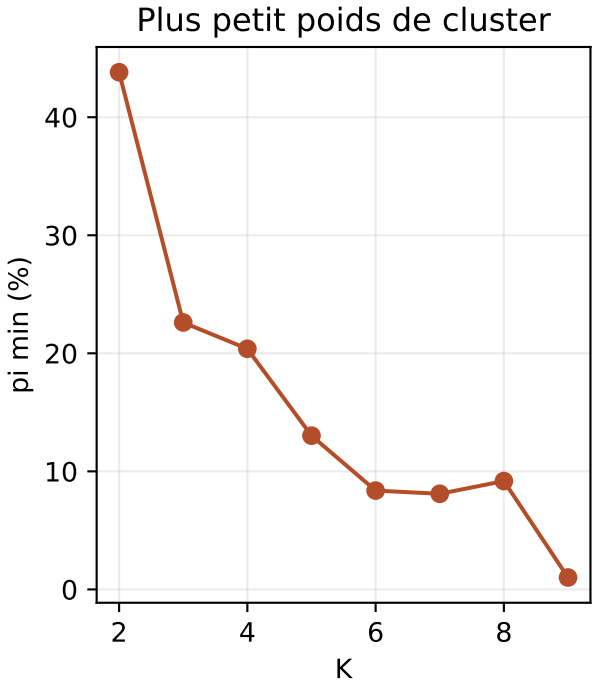




K	alpha0 min	alpha0 med.	alpha0 max	pi min %	pi med. %	pi max %
2	9.18	14.88	20.59	37.9	50.0	62.1
3	6.62	16.72	19.98	25.4	35.8	38.8
4	9.04	17.57	29.54	20.6	24.5	30.4
5	8.73	21.23	29.16	7.2	20.7	30.1
6	7.86	19.53	30.09	7.0	17.7	24.0
7	7.79	20.89	32.08	2.3	16.2	19.1
8	8.31	20.75	33.01	2.4	14.2	19.3
9	8.07	20.73	35.04	2.5	11.4	19.5

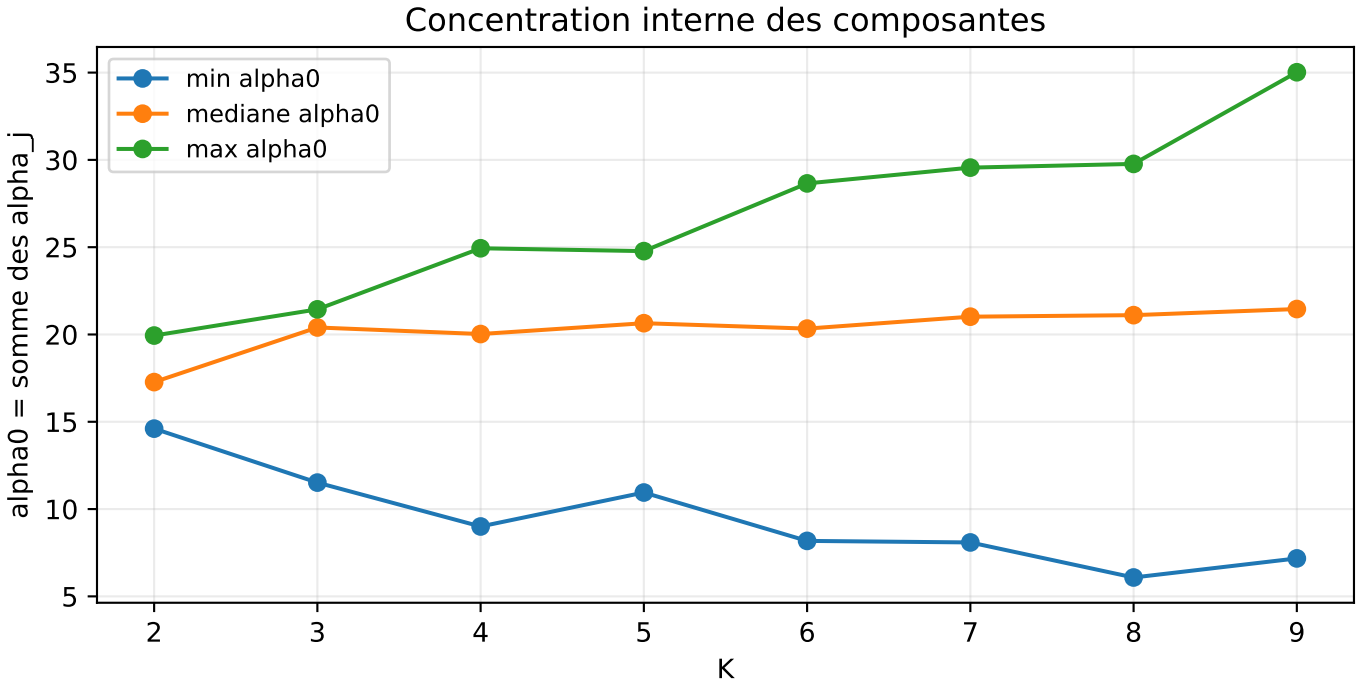
DMM - Diagnostic alpha0 - 2014 - 9 postes

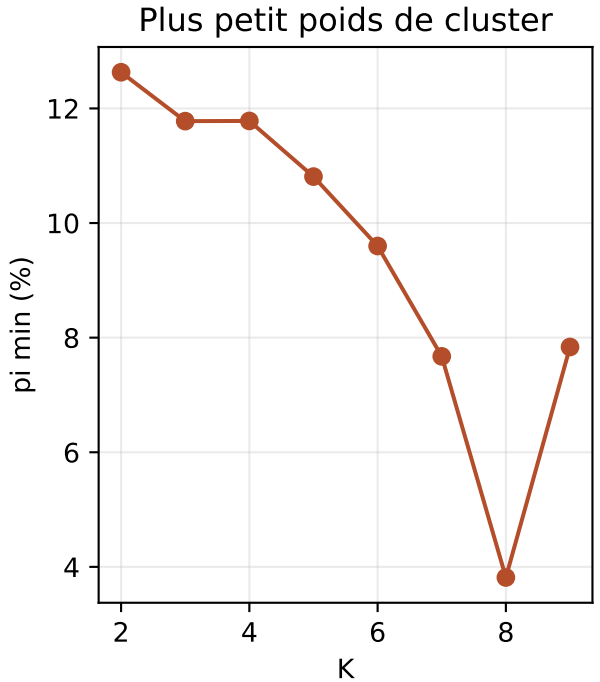




K	alpha0 min	alpha0 med.	alpha0 max	pi min %	pi med. %	pi max %
2	9.75	15.03	20.30	43.8	50.0	56.2
3	8.04	19.91	23.38	22.6	35.8	41.6
4	8.62	16.89	23.94	20.4	21.5	36.7
5	8.35	17.31	28.29	13.0	20.6	25.9
6	9.22	21.96	28.40	8.4	15.4	25.4
7	7.30	18.16	27.91	8.1	13.7	23.4
8	7.74	18.91	29.92	9.2	11.9	20.4
9	7.91	20.04	29.71	1.0	11.6	20.7

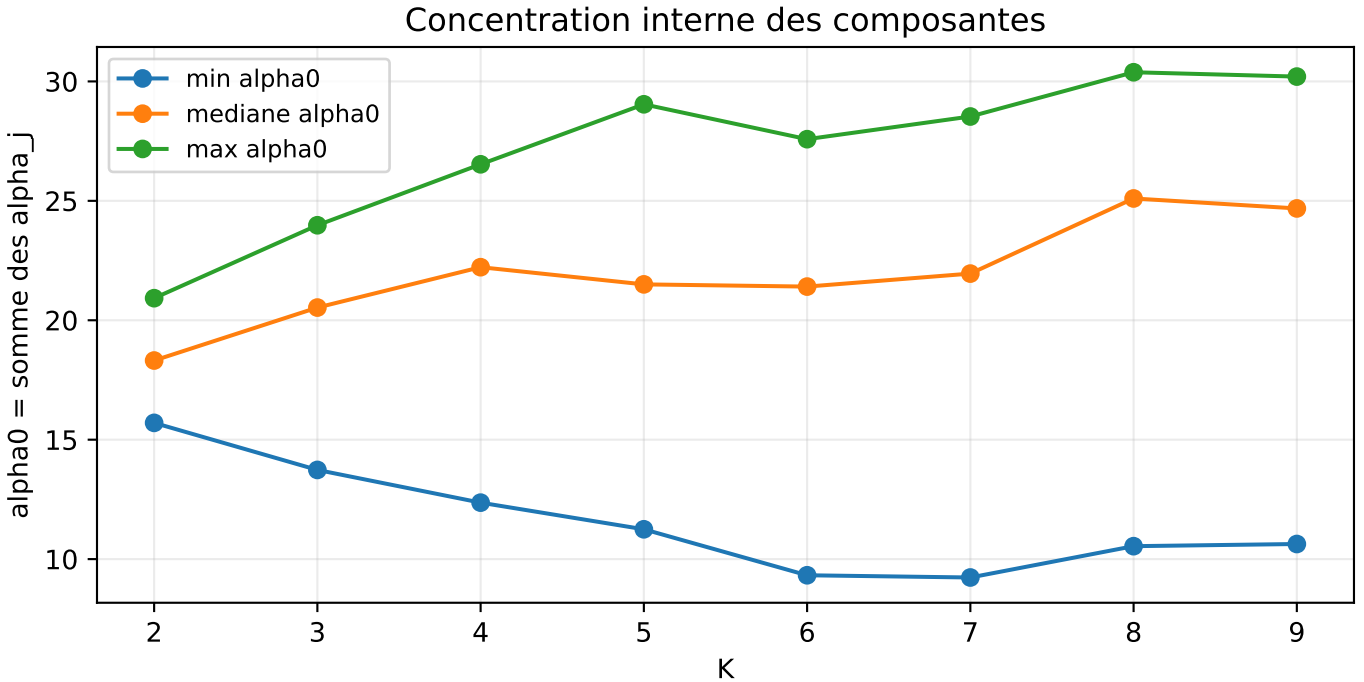
DMM - Diagnostic alpha0 - 2007 - 23 postes

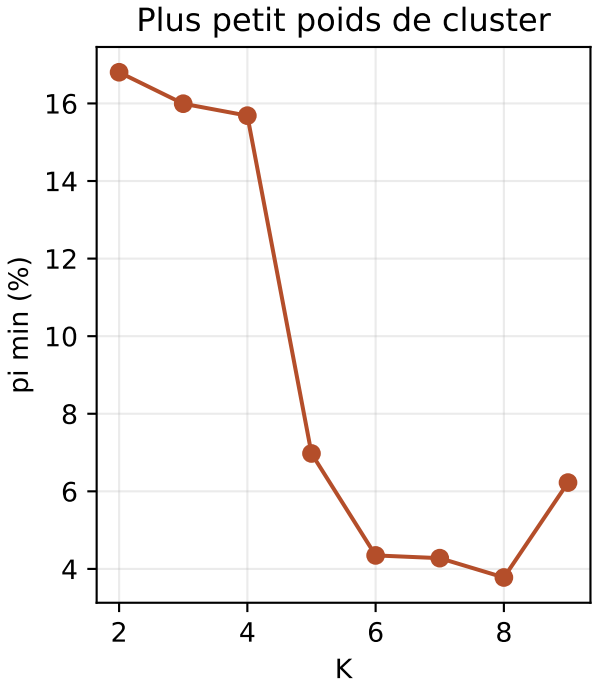




K	alpha0 min	alpha0 med.	alpha0 max	pi min %	pi med. %	pi max %
2	14.61	17.27	19.94	12.6	50.0	87.4
3	11.52	20.40	21.44	11.8	38.4	49.8
4	9.00	20.03	24.94	11.8	15.6	56.9
5	10.95	20.65	24.77	10.8	21.6	29.4
6	8.18	20.34	28.65	9.6	14.8	24.8
7	8.09	21.02	29.56	7.7	11.2	23.2
8	6.08	21.11	29.77	3.8	11.6	20.3
9	7.17	21.46	35.02	7.8	10.6	17.9

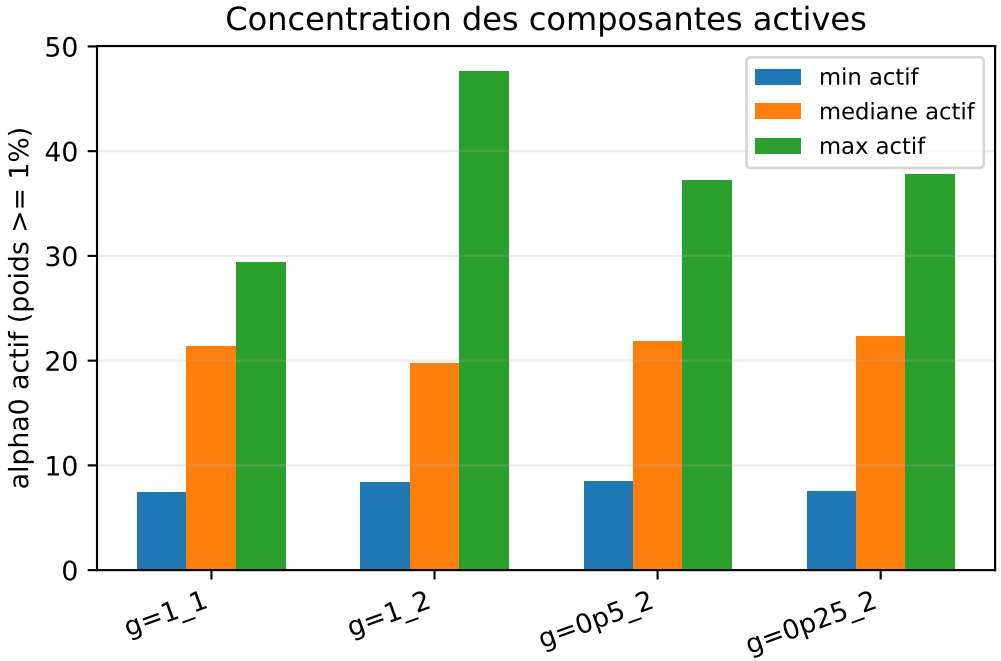
DMM - Diagnostic alpha0 - 2014 - 23 postes

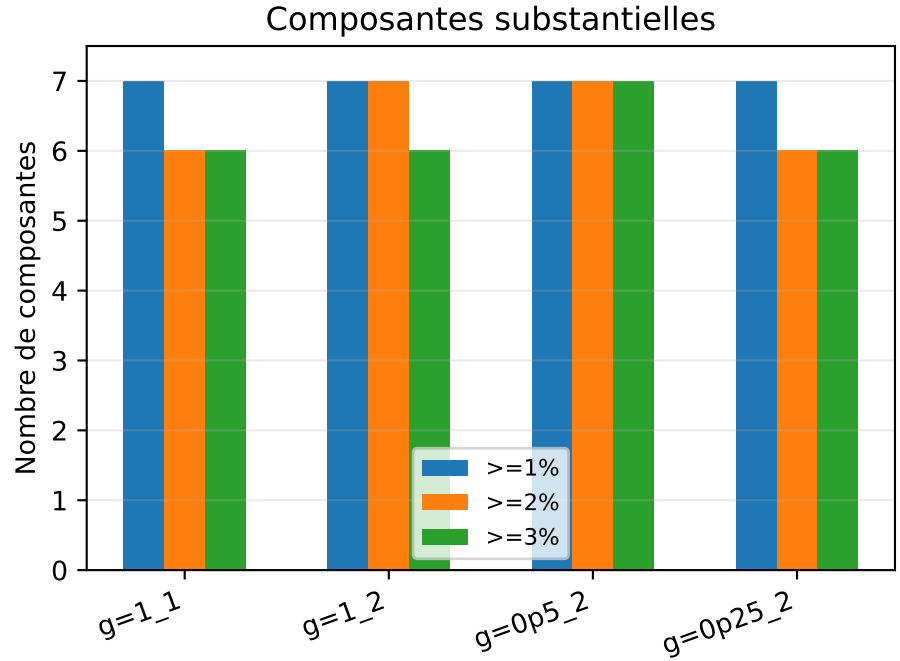




K	alpha0 min	alpha0 med.	alpha0 max	pi min %	pi med. %	pi max %
2	15.70	18.31	20.92	16.8	50.0	83.2
3	13.73	20.53	23.98	16.0	39.8	44.2
4	12.36	22.22	26.53	15.7	25.3	33.7
5	11.25	21.50	29.04	7.0	21.6	31.6
6	9.32	21.41	27.58	4.3	16.7	26.4
7	9.23	21.95	28.53	4.3	13.8	22.8
8	10.54	25.10	30.38	3.8	12.4	18.1
9	10.63	24.68	30.20	6.2	11.7	16.9

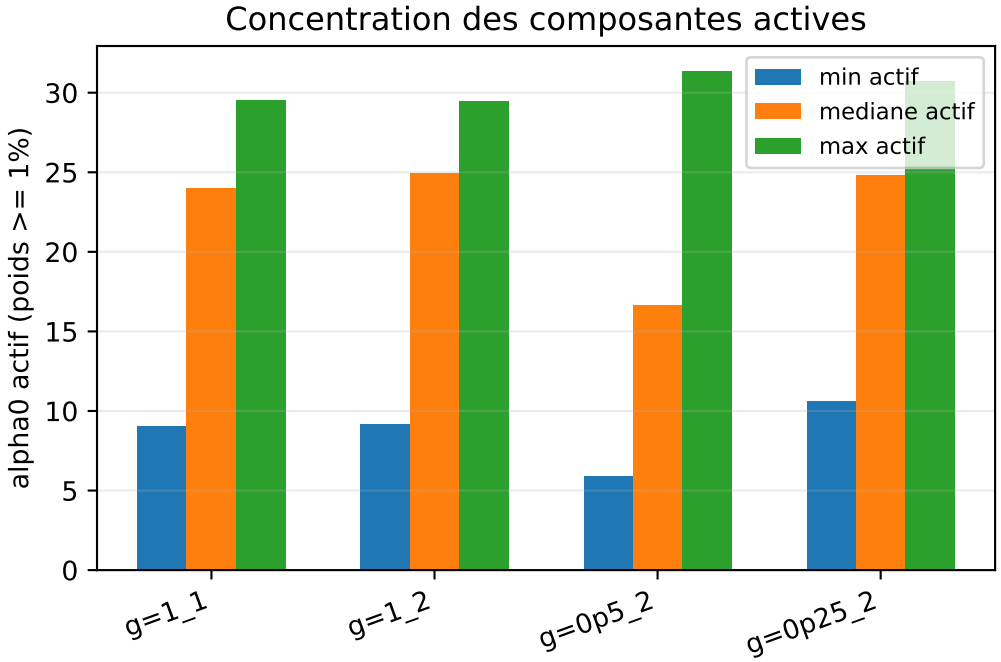
DPMM - Diagnostic alpha0 et poids - 2007 - 9 postes

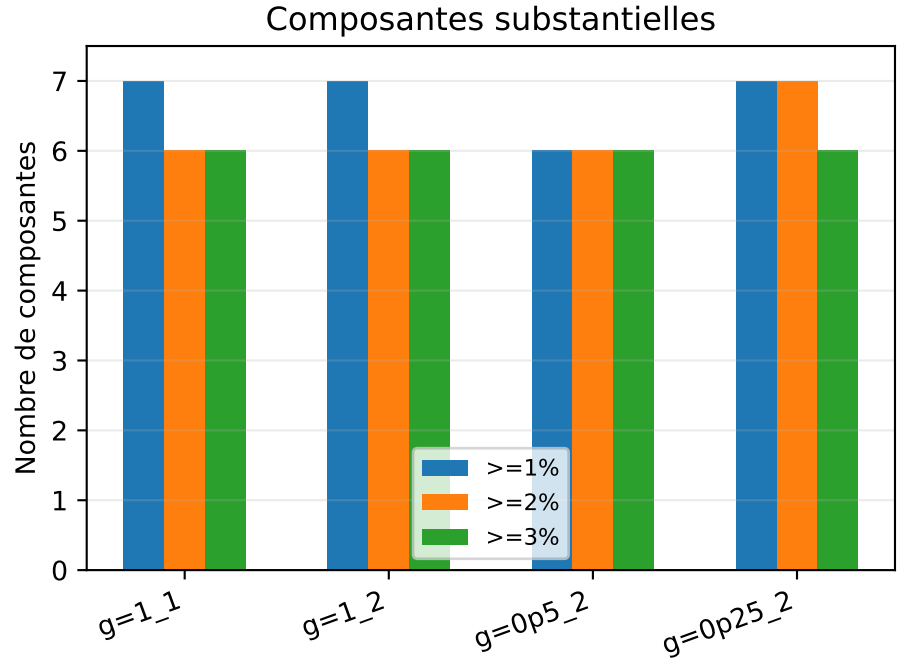




Prior	Hyperparametres	act. >=1%	alpha0 min	alpha0 med.	alpha0 max	poids min %	poids max %
gamma_1_1	Gamma(a=1.0, b=2.0)	7	7.47	21.34	29.43	1.02	33.8
gamma_1_2	Gamma(a=0.25, b=2.0)	7	8.41	19.75	47.66	2.04	31.3
gamma_0p5_2	Gamma(a=0.1, b=2.0)	7	8.51	21.84	37.17	3.66	30.6
gamma_0p25_2	Gamma(a=0.05, b=2.0)	7	7.54	22.35	37.76	1.02	31.6

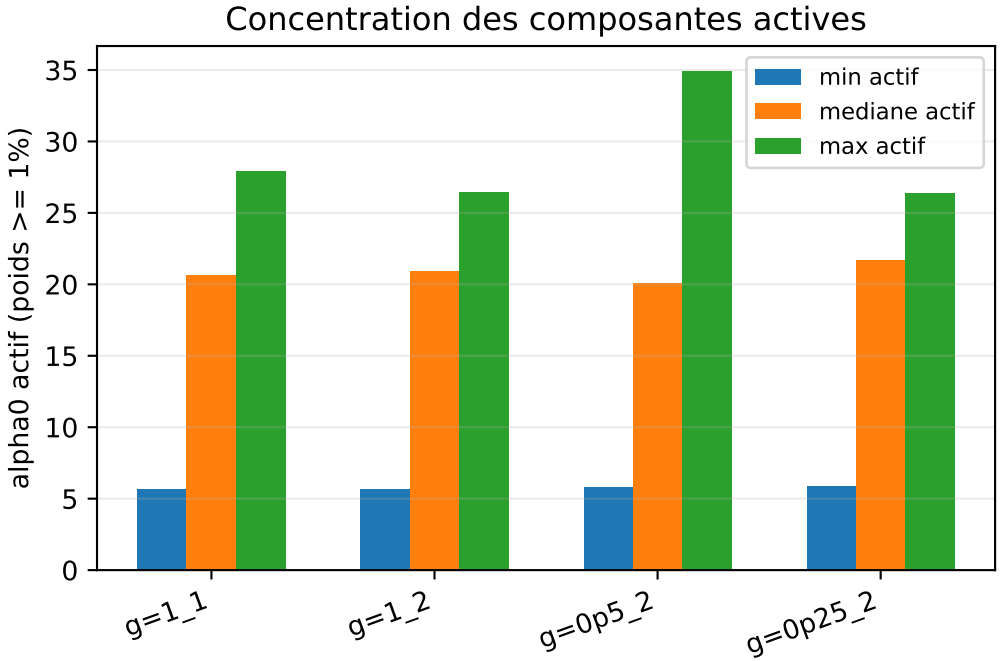
DPMM - Diagnostic alpha0 et poids - 2014 - 9 postes

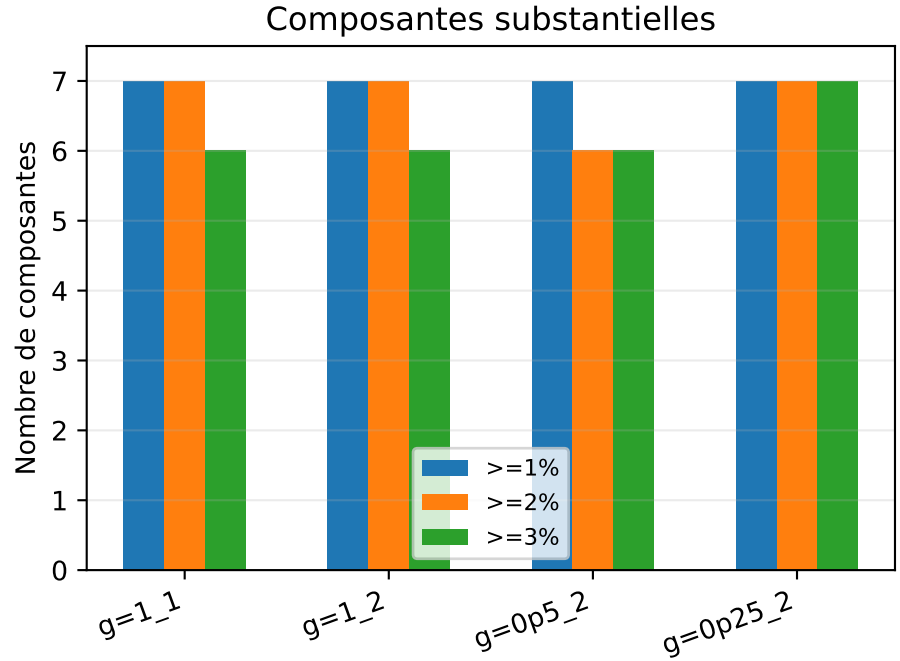




Prior	Hyperparametres	act. >=1%	alpha0 min	alpha0 med.	alpha0 max	poids min %	poids max %
gamma_1_1	Gamma(a=1.0, b=2.0)	7	9.02	23.96	29.54	1.05	38.1
gamma_1_2	Gamma(a=0.25, b=2.0)	7	9.19	24.96	29.47	1.52	38.1
gamma_0p5_2	Gamma(a=0.1, b=2.0)	6	5.87	16.63	31.37	0.98	27.4
gamma_0p25_2	Gamma(a=0.05, b=2.0)	7	10.60	24.78	30.71	2.59	38.8

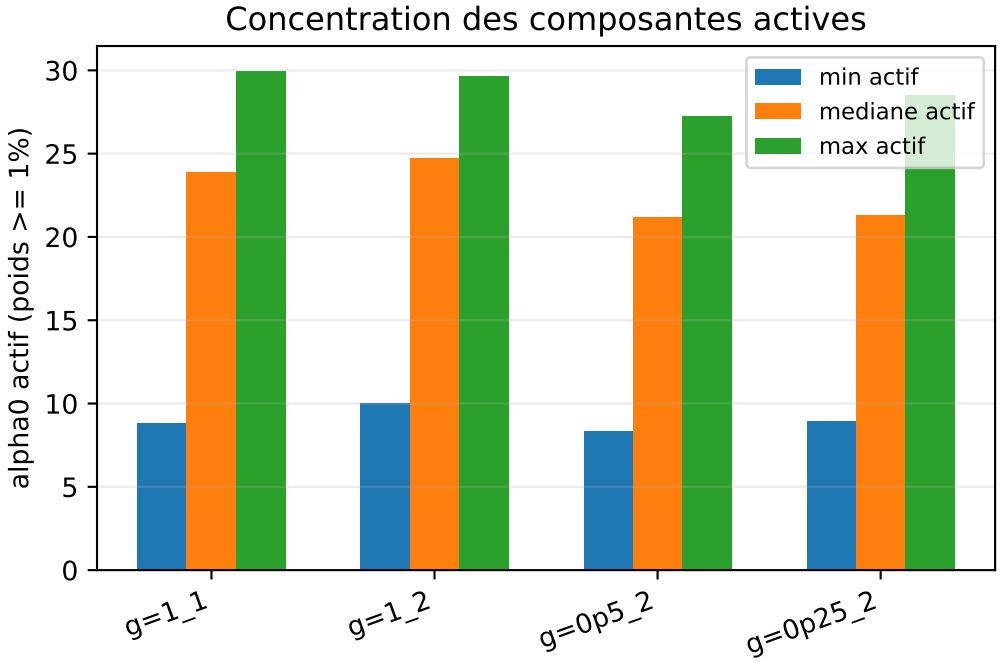
DPMM - Diagnostic alpha0 et poids - 2007 - 23 postes

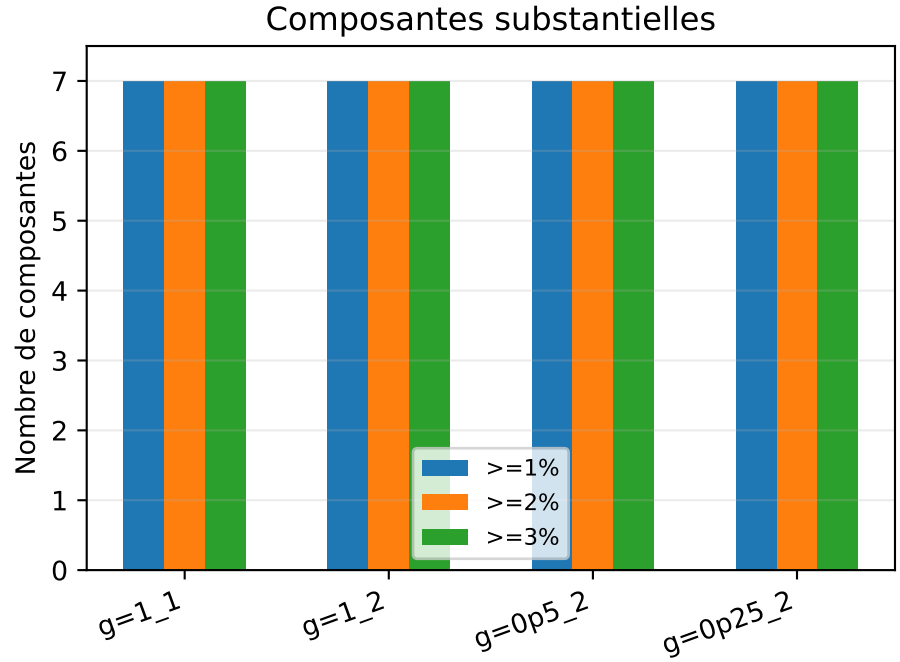




Prior	Hyperparametres	act. >=1%	alpha0 min	alpha0 med.	alpha0 max	poids min %	poids max %
gamma_1_1	Gamma(a=1.0, b=2.0)	7	5.65	20.62	27.93	2.63	30.4
gamma_1_2	Gamma(a=0.25, b=2.0)	7	5.64	20.89	26.47	2.74	26.2
gamma_0p5_2	Gamma(a=0.1, b=2.0)	7	5.82	20.10	34.93	1.71	33.6
gamma_0p25_2	Gamma(a=0.05, b=2.0)	7	5.83	21.70	26.39	3.38	31.4

DPMM - Diagnostic alpha0 et poids - 2014 - 23 postes





Prior	Hyperparametres	act. >=1%	alpha0 min	alpha0 med.	alpha0 max	poids min %	poids max %
gamma_1_1	Gamma(a=1.0, b=2.0)	7	8.79	23.86	29.96	3.97	28.6
gamma_1_2	Gamma(a=0.25, b=2.0)	7	10.03	24.69	29.66	3.56	31.0
gamma_0p5_2	Gamma(a=0.1, b=2.0)	7	8.32	21.19	27.24	7.26	23.0
gamma_0p25_2	Gamma(a=0.05, b=2.0)	7	8.90	21.31	28.52	5.72	20.3